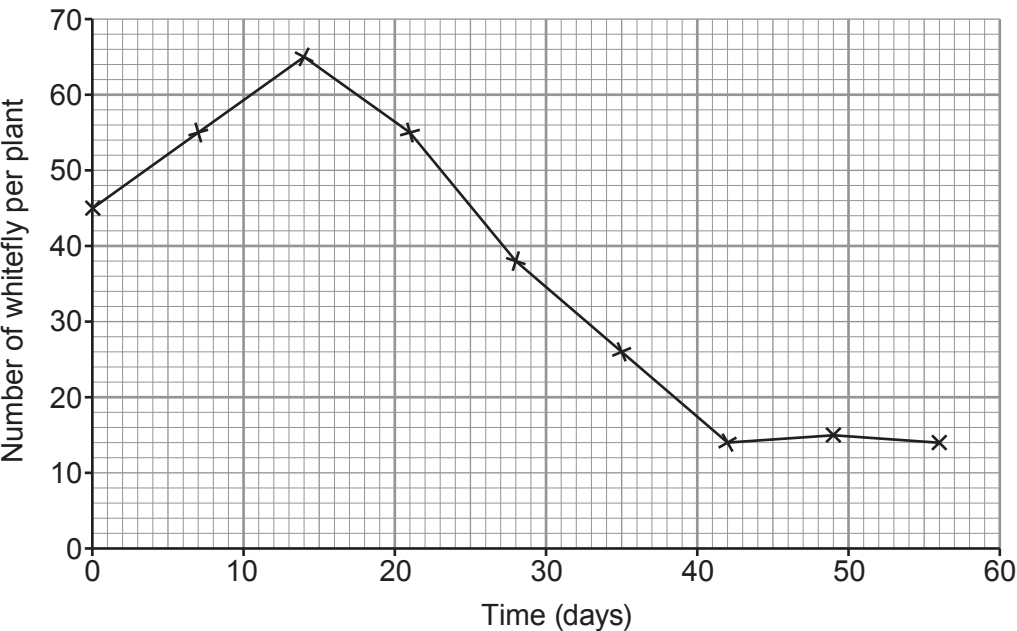


Graph 1.1 shows the number of whiteflies in a greenhouse containing tomato plants. *E. formosa* were introduced on day 7.

Graph 1.1



- (i) I. The use of *E. formosa* to reduce the number of whiteflies is considered to be successful when there are 20 or fewer whiteflies per plant. Use **Graph 1.1** to determine how many days it took for the number of whiteflies to fall to 20 following the introduction of *E. formosa*. [2]

..... days

- II. Suggest a reason why it took this long for the number to fall to 20. [1]

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- (ii) Suggest **one** reason why this method of pest control would not be effective to use if whiteflies damaged wheat crops grown in Wales. [1]

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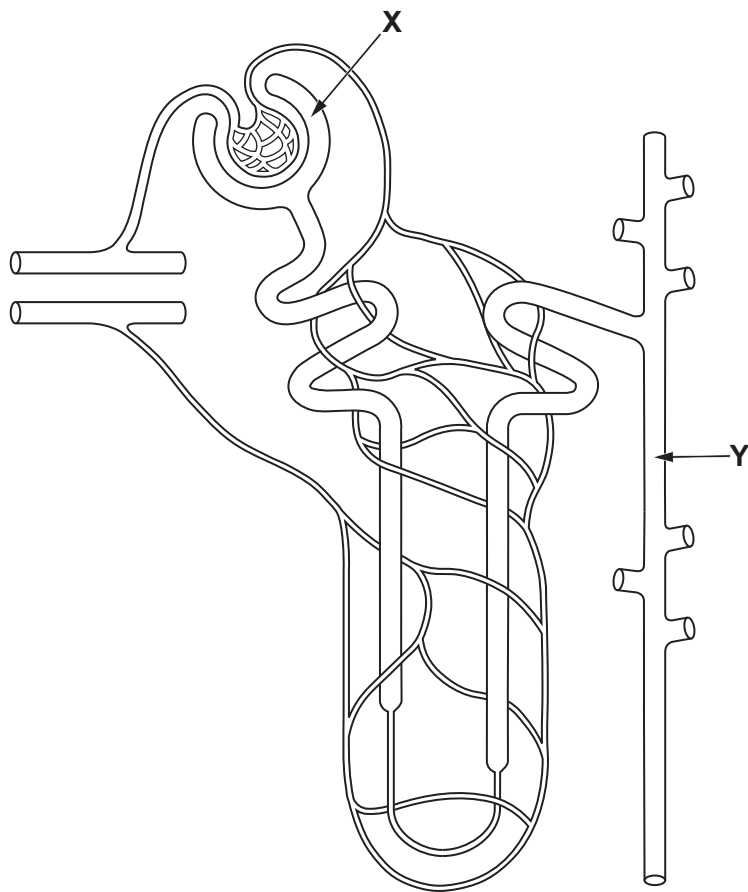
- (iii) An alternative approach to reducing pest numbers is to use pesticide. State why it is not appropriate to use pesticide along with *E. formosa*. [1]

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7. **Image 7.1** shows the structure of a nephron.

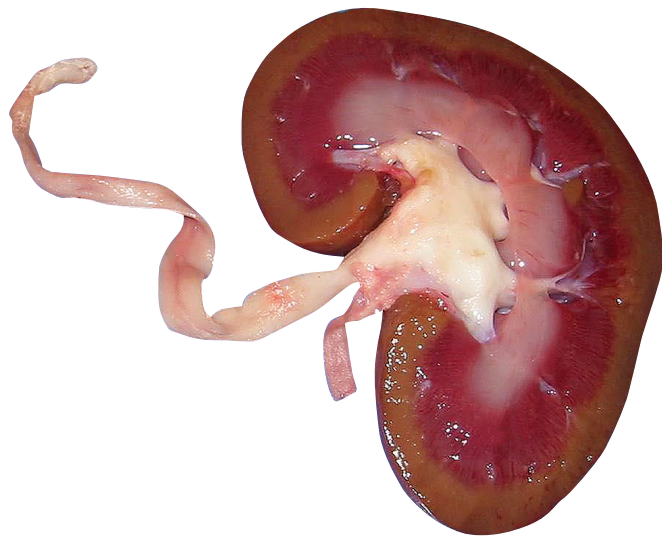
Image 7.1



- (a) **Use a labelled line** to show the position of structure **X** (shown in **Image 7.1**) in the section through the kidney in **Image 7.2**.

[1]

Image 7.2



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only

- (b) The filtrate found at **X** in **Image 7.1** differs from the filtrate found at **Y** in **Image 7.1**. Describe and explain the composition of the filtrate at **X** and **Y**, identifying structures **X** and **Y** in your answer.
(No reference to ADH is required) [6 QER]

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- (c) Protein is found in the blood in the capillary knot but is not found in the filtrate at **X**. Explain this observation. [2]

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